Slovak University of Technology in Bratislava
Faculty of Informatics and Information Technologies

Methods of Engineering Work

Academic Year 2026/2027

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Application for Applied Research (Call VV-2024)

Item VV-A1-03: Project Title

Implementation of Post-Quantum Cryptography in V2X Systems

Item VV-A1-04: Project Acronym

V2X-PQC

Item VV-A1-09: Project Annotation

The advent of quantum computers poses an imminent security threat to modern cooperative intelligent transportation systems, making the transition to post-quantum cryptographic (PQC) standards necessary. Although the standardized ML-DSA (Dilithium) algorithm is widely applicable in the IT sector, it faces significant limitations in vehicle-to-everything (V2X) communication. Its 2,420-byte digital signature can overload communication channels under high traffic density. The alternative Falcon algorithm, with a 666-byte signature, reduces bandwidth consumption but entails substantial computational costs for verification. At busy intersections, this places an excessive burden on roadside units (RSUs), causing the critical latency threshold of 10 ms to be exceeded.

The main objective of the V2X-PQC project is to investigate and optimize PQC algorithms for V2X communication and develop a specialized software solution for roadside units (RSUs) and vehicle on-board units (OBUs). The proposed system will enable highly efficient verification of cryptographic signatures on messages shared among road users through PQC header compression and multithreaded processing of standardized Basic Safety Messages (BSMs).

The key project outcome will be an IEEE 1609.2-compatible software solution that reduces the volume of transmitted cryptographic data by more than 40% and guarantees verification latency below 10 ms even at a load of 1,000 messages per second. Through hardware-specific code optimization, the resulting software will achieve a throughput of over 1,200 PQC signatures per second and reduce latency to below 5 ms. The project will optimize existing security standards for connected vehicles and adapt them to current technical requirements in the automotive industry.

VV-C – Budget of the Project in EUR

Applicant

Budget category	2027 (EUR)	2028 (EUR)	Total (EUR)
01. Direct running costs	32,619.20	33,019.20	65,638.40
02. Wage and other personal costs	21,600.00	21,600.00	43,200.00
03. Social and health insurance	7,819.20	7,819.20	15,638.40
04. Travel costs	500.00	1,500.00	2,000.00
05. Material	1,500.00	900.00	2,400.00
06. Services	900.00	900.00	1,800.00
07. Energy, water, communications	300.00	300.00	600.00
08. Popularization costs	0.00	0.00	0.00
09. Indirect costs	1,500.00	1,500.00	3,000.00
10. Total running costs	**34,119.20**	**34,519.20**	**68,638.40**
Total costs from APVV	**34,119.20**	**34,519.20**	**68,638.40**
Financing from other sources	0.00	0.00	0.00
State (outside APVV)	0.00	0.00	0.00
Foreign	0.00	0.00	0.00
Private	0.00	0.00	0.00
Total costs	**34,119.20**	**34,519.20**	**68,638.40**

List of Expected Costs Covered by APVV

No.	Expected costs
02	**Wage and other personal costs – 43,200.00 EUR** Principal Investigator: 500 hours/year at 18 EUR/hour (18,000 EUR in total); Cryptography Researcher: 300 hours/year at 16 EUR/hour (9,600 EUR); Software/V2X Developer: 300 hours/year at 14 EUR/hour (8,400 EUR); Testing and Evaluation Researcher: 300 hours/year at 12 EUR/hour (7,200 EUR).
03	**Social and health insurance – 15,638.40 EUR** Employer contributions estimated at 36.2% of gross wages: 43,200.00 EUR $\times$ 0.362 = 15,638.40 EUR.
04	**Travel costs – 2,000.00 EUR** 2027: domestic research meetings and technical workshop (500 EUR). 2028: one scientific conference or specialist workshop, travel and accommodation (1,500 EUR).
05	**Material – 2,400.00 EUR** 2027: development peripherals, network interfaces, cables and laboratory consumables (1,500 EUR). 2028: adapters, replacement components and additional experimental materials (900 EUR).
06	**Services – 1,800.00 EUR** 2027: computing services, hosting and technical support (900 EUR). 2028: computing resources, storage and technical services (900 EUR).
07	**Energy, water, communications – 600.00 EUR** Electricity and network communications related to testing and computations: 300 EUR per year.
08	**Popularization costs – 0.00 EUR** No funds requested.
09	**Indirect costs – 3,000.00 EUR**

No.	Expected costs
	Administrative, accounting and shared laboratory support: 1,500 EUR per year.

Summary Budget of the Project

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VV-D – Project Schedule

Begin of phase	End of phase	Phase name
01.01.2027	30.04.2027	Analysis of V2X security requirements and post-quantum signature algorithms
01.05.2027	31.10.2027	System architecture design and development of the initial PQC software prototype
01.11.2027	30.04.2028	Integration of cryptographic mechanisms into the simulated V2X environment
01.05.2028	30.09.2028	Performance testing and optimization of signature verification and communication overhead
01.10.2028	31.12.2028	Final validation, evaluation of results and technical documentation

List of Devices in Project

Name	Description
Research workstations	Computers for software development, V2X simulation and cryptographic benchmarking.
Network testing components	Network interfaces, adapters and cables for experimental communication tests.
Embedded development modules	Small-scale development boards for performance measurements of cryptographic implementations.
Computing infrastructure	Computing and storage resources for simulations, processing test data and evaluating results.